Interface Control Document

SANT-20-00023

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Revision: E

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1 Document Control

Revision	Date	Section	Description
D	2018-01-26	All	First Release
E	2019-06-04	4, 5, 6	Second Release

2 Revision Control

Product	Part Number	Revisions Covered	Notes
SANT	SANT-20-00023	K	

3 Related Documents

No.	Document Name	Document Reference
OPT-20-00023	Options Sheet: SANT	Rev A

4 Overview

The CPUT S-band antenna (SANT) is a circularly polarised all metal patch antenna designed for use within the Amateur-Satellite Service. The SANT conforms to the CubeSat standard and is thus primarily designed for CubeSat missions but can be used on larger satellites. The SANT is compatible with CPUT S-band Transmitter (STX), which is used in telemetry or payload data downlinks.

5 Features

- Left or Right Hand Circular Polarization (Default Left)
- Covers the Amateur band in the S-band frequency range (2.4 to 2.45 GHz)
- Simple integration with structure
- High Gain (>7 dBi)
- SMA connector (straight or right angle)

6 Detailed Characteristics

The SANT has been optimised to be mounted in the centre of a 3U CubeSat as depicted in Figure 1. The measured performance results in this document are based on this configuration so deviations from this intended usage might result in less than optimum performance.

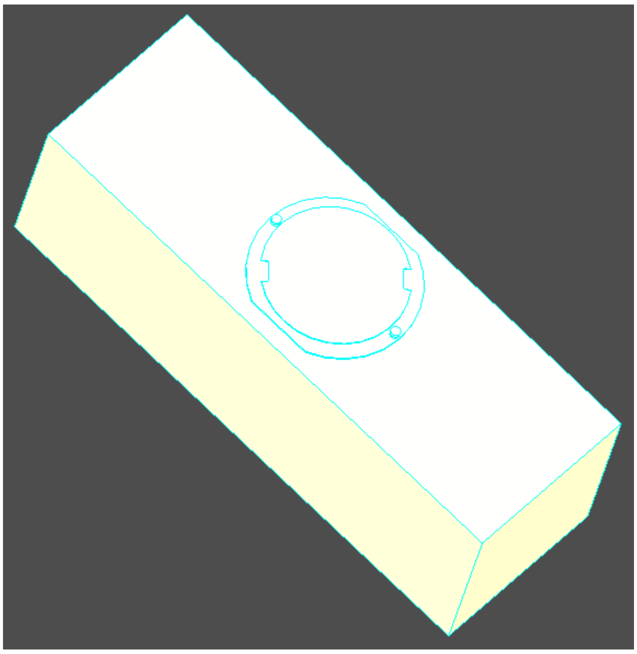

Figure 1: CAD model of SANT mounted in the centre of a 3U CubeSat

6.1 Typical Performance

The SANT is fully characterised within an anechoic chamber by subjecting it to comprehensive testing of various performance parameters. The particular parameters measured are Input Reflection Coefficient (S_{11}), maximum gain, radiation patterns, and axial ratio. Patterns are derived from S21 measurements between the AUT and a linearly polarised horn antenna as reference antenna. Two pattern cuts are measured; one with the reference antenna in the horizontal orientation (0°) and the other in the vertical (90°) orientation. Below are some typical measured results. The results of the performance parameters displayed below will vary between each manufactured unit due to manufacturing tolerances.

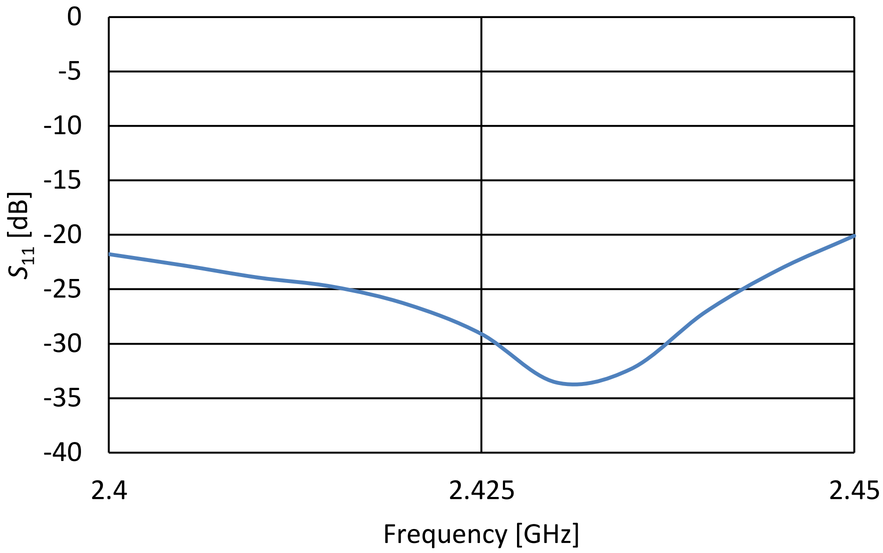

Figure 2: Typical Input Reflection Coefficient (S_{11}) Measurement

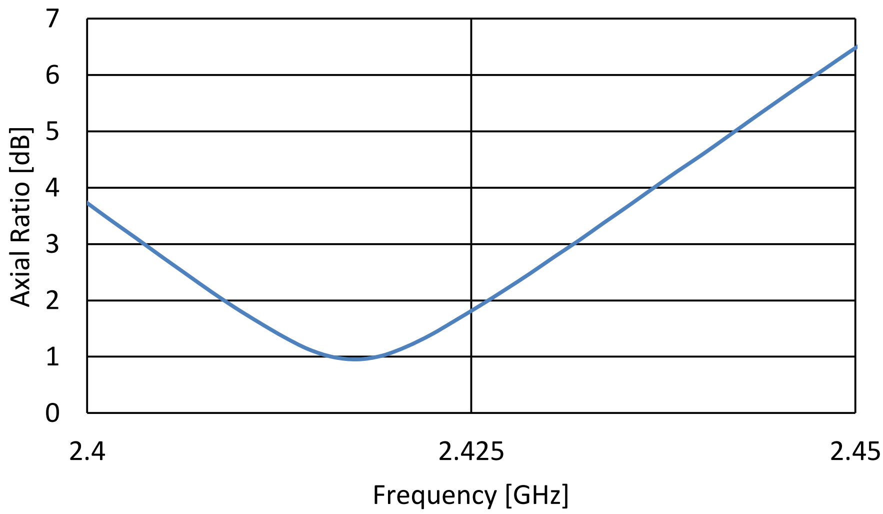

Figure 3: Typical Axial Ratio Measurement

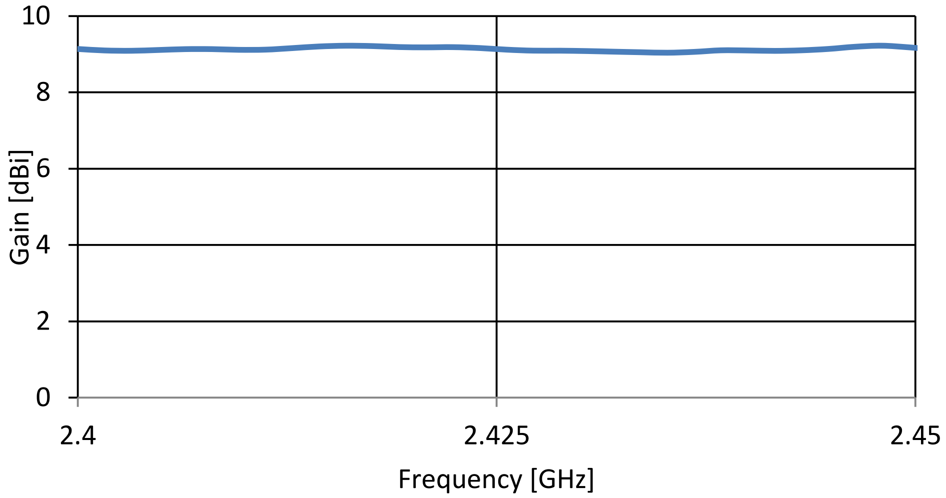

Figure 4: Typical Gain Measurement

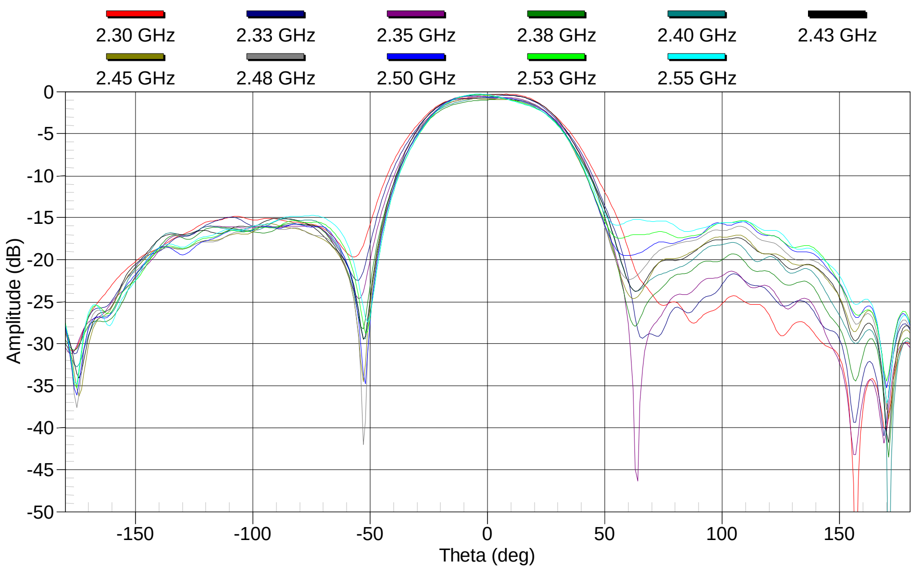

Figure 5: Typical Far field Pattern – SANT 0°, Horn Antenna 0°

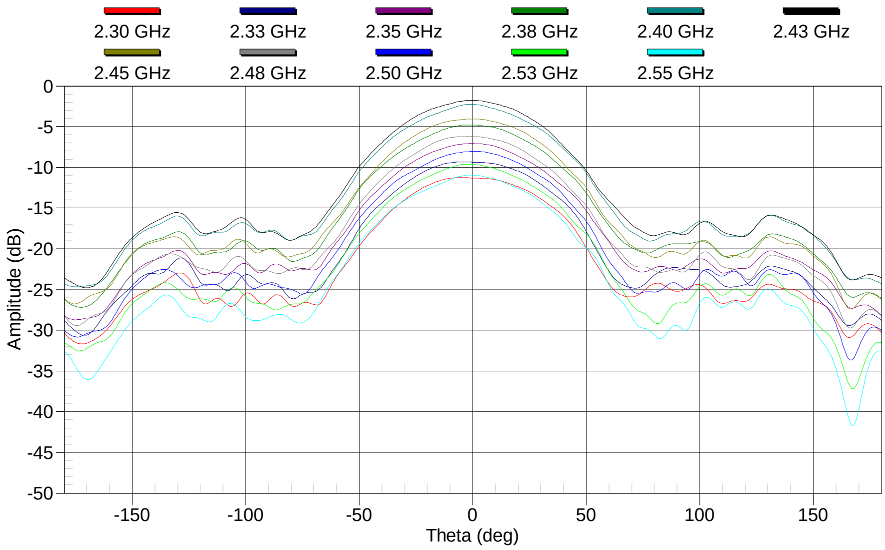

Figure 6: Typical Far Field Pattern – SANT 0°, Horn Antenna 90°

The half power beamwidth of the SANT is approximately 60 degrees within its operating frequency range.

6.2 Mechanical Specifications

6.2.1 Physical Specifications

- Mass < 50g (including connector)
- Dimensions (Figures 7, 8 and 9)
  – 89 mm x 81.5 mm x 4.1 mm
  – Maximum height above the CubeSat chassis: 4.1 mm
  – The SMA connector intrudes into the chassis by 10 mm with the straight SMA (11 mm with right angled SMA)

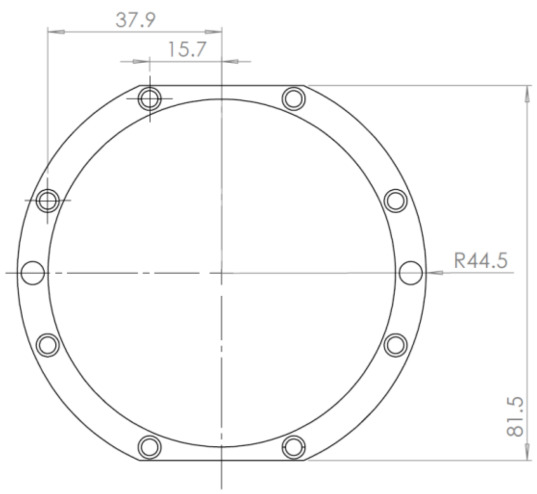

Figure 7: SANT Dimensions – Viewed from above

6.2.2 Interfaces

- Attachment – 8 x M2.5 counter sunk mounting screws
- RF connector – 50 Ohm SMA female connector: Straight (default) or Right Angled (optional)

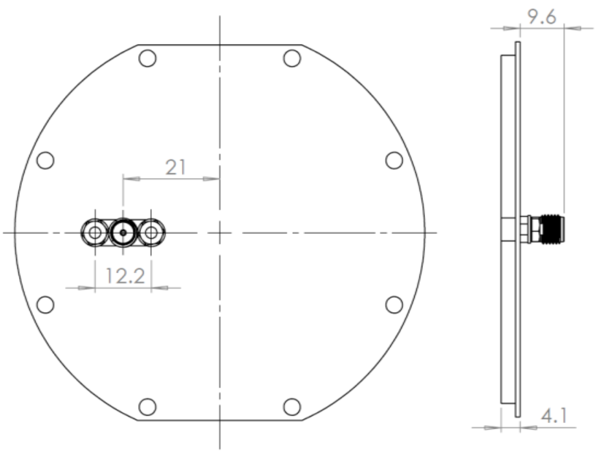

Figure 8: SANT Dimensions- Viewed from below and from the side with a straight SMA connector

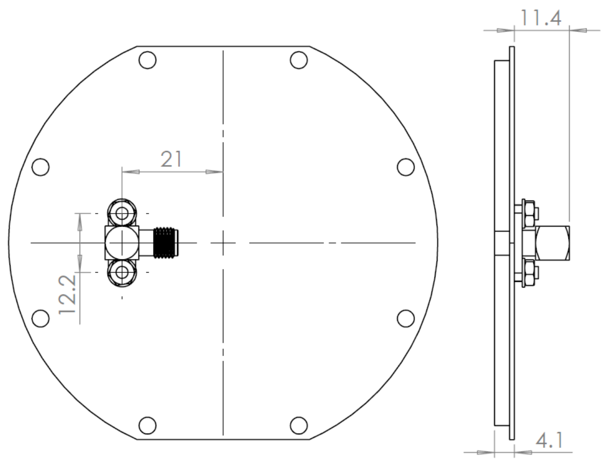

Figure 9: SANT Dimensions – Viewed from below and from the side with a right angled SMA connector

6.3 Recommended mounting instructions

The patch antenna is optimised to be ideally placed at the centre of a 3U structure; mounting the antenna in a different position does not guarantee the same performance results. The antenna is to be mounted on top of, not under, a ground plane which is typically electrically tied to the satellite chassis (ground). The ground plane must be an electrically conductive surface; failing to do so may result in a significantly degraded performance as well as affect the match to the power amplifier of a transmitter. Note that an anodised metallic surface does not provide the required conductive surface; instead a chromate conversion coating is suggested to maintain electrical conductivity. Furthermore, countersunk metallic screws must be used to ensure that fixtures do not extend above the ground plane of the antenna whilst maintaining electrical contact. Plastic screws may be a viable alternative if fasteners cannot be countersunk flush with the antenna's base plate. Antenna performance is influenced by objects in its proximity, thus avoid placement of any components near the vicinity (within 2 cm) of the radiating element. Consult F'SATI or its distributors if deviating from the recommended mounting instructions.